Drag & Speak: A Multimodal Audio-haptic Approach for Arithmetic Operations for Screenreader Users

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Abstract. Drag & Speak assists people who are blind or have low vision in learning and performing basic arithmetic calculations. It supports tactile exploration of digits and monitors an index finger. The setup consists of a camera and a cut out wooden plate for tactile support when placing digits into the plate's grooves. Each digit is a small tangible brick-like object consisting of an ArUco marker and Braille label. A user can speak to a voice assistant and receive support when navigating to a cell in the grid by sonification and also for simple one digit arithmetic calculations. A user study in a within-subjects design with three experimental conditions for 8 participants shows good SUS values and acceptable TLX scores.

Keywords: Multimodal · Tangible · Arithmetic.

1 Introduction

Many STEM (Science, Technology, Engineering and Mathematics) disciplines are dependent on non-linear notations, as characters and symbols are combined with graphics to write formulae and develop diagrams. Already basic arithmetic operations can be challenging to learn for students who are blind or have low vision. Addition, subtraction, multiplication, and division require placing digits and numbers both horizontally and vertically aligned to each other when reading and writing calculations.

Although screenreaders can indicate digits verbally, it is cumbersome to explore numbers spatially by text-to- speech output, in particular as empty space is neglected by screenreaders. As an alternative, mathematical braille and braille displays are suitable to write and read the terms needed including spaces. Still, braille displays are not presenting a two-dimensional layout. If students fail to learn how to explore digits horizontally and vertically, their success in mastering STEM subjects can be limited.

Other non-visual approaches are needed to combine both visual and non-visual design of interactive systems for learning how mathematical tasks can be solved. We present a novel multimodal approach to basic arithmetic operations in a tabular layout by integrating voice input with placement of tangibles and providing feedback to the student with text-to-speech output as well as sonification. Such an approach may be extended to more complex notations and diagramming techniques.

2 Related Work

Both tactile as well as acoustic modalities can contain information about two-dimensional layout. Spatial sound can present spoken text in a horizontal layout if Header Related Transfer Functions (HRTFs) are implemented [4] but vertical resolution is rather limited and depends on individual preferences. Large pin array displays [11] can hold only a few lines with only some characters. The Hyperbraille display shows 12 lines with 40 characters each but is not easily affordable. [10] have developed one of the first eLearning systems to teach geometrical concepts on a tactile display with 7200 pins.

Low-cost physical bricks can carry braille labels and allow for tactile interaction, if accidental movements can be avoided. The Brannan Cubarithm Kit Slate 100 Cubes [8] allows placement of the braille labeled cubes in a tabular rubber-based frame.

Breiter et. al. suggest also grooves in a plate to stabilize the position of wooden blocks [3]. In this work the concept of tangible objects is applied to allow digitalisation of the editing process. The Tangible Grid [7] consists of a tactile grid layout to design web pages by placing physical objects.

In a digital system individual feedback can be generated to support learners. Ali et. al. have developed a system guiding students who are blind or have low vision through structural information in a graphical user interface and a screenreader for solving mathematical tasks [1]. They show an increase of efficiency but do not focus on effectiveness and learning.

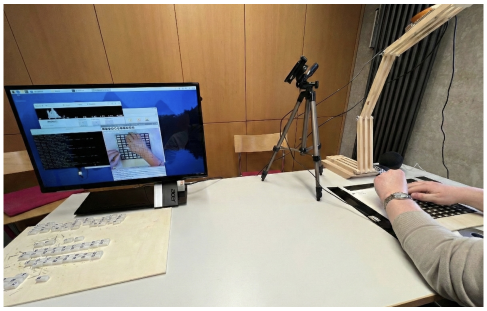
The Tactonom [6] is built around affordable components: tactile feedback from microcapsule paper, a camera mounted above and a casing with a small computer generating text-to-speech output for regions identified through a finger. Similar audiohaptic devices are Talking Tactile Tablet and IVEO, but these rely on more expensive touch input sensing devices.

In this work, we extend the concept of a Tactonom by introducing tangible objects and real-time hand tracking as described below, implement sonification for guiding fingers [9] and evaluate the system with students who are blind or have low vision.

3 Methodology

3.1 System Implementation

To achieve a low-cost, replicable, and portable solution, our prototype is built around a Raspberry Pi 4 as a standalone computational unit. As shown in Fig. 1(a), the setup includes a custom height-adjustable wooden stand with a top-down Arducam camera to keep the tactile grid within the camera's field



(a) Setup of the prototype

(b) Close-up view of the tactile grid and tangibles

Fig. 1. System overview. (a) Prototype setup with the custom wooden camera stand and the experimenter's monitor showing real-time object detection and hand tracking. (b) Close-up view of a user exploring the tactile grid and placing 3D-printed tangibles.

of view. The system further includes a speaker, microphone, tactile grid, and tangibles.

The software architecture is implemented in Python. The vision module uses OpenCV to detect the grid structure and multiple ArUco markers [5] in real time, including their IDs and positions. Concurrently, the MediaPipe framework is integrated for Hand Landmark Detection, fingertip tracking for deictic interaction, and recognition of the hand region boundary. This allows the system to identify the pointed grid cell and distinguish temporary palm occlusion from deliberate object removal. The recognition feedback is visualized in real time in the monitor view shown in Fig. 1(a).

For voice interaction, we combine online Automatic Speech Recognition (ASR) with a local offline Text-to-Speech (TTS) engine to balance accuracy and latency. Early tests with local ASR solutions (Whisper and Vosk) on the Raspberry Pi 4 did not provide sufficient real-time performance or recognition accuracy. We therefore used AssemblyAI for streaming ASR, local command parsing, and Piper for offline TTS. We also shortened prompts and feedback based on pilot testing to reduce waiting time and improve interaction fluency.

3.2 Tangible and Grid Design

The physical layer underwent several design iterations to optimize the haptic experience and recognition efficiency.

Tangibles We replaced off-the-shelf solutions (e.g., Lego or wooden blocks) with custom 3D-printed tangibles. This approach reduced weight and allowed for precise sizing to match the ArUco markers. Because the camera must capture the whole grid from above, marker size and image resolution must be balanced against available cell space and processing load: larger markers reduce usable cell space, while higher resolution increases computation on the Raspberry Pi.

In early design stages, we tested several QR-based options, including standard QR codes with different error-correction levels and Micro QR codes. In practice, they were either too slow for real-time multi-object detection at small sizes or unsuitable for integrating Braille information. Based on these tests, we finally adopted ArUco markers for tangible recognition. Unlike QR codes, which are designed to encode larger amounts of information, the markers in our system only need to distinguish between different objects. ArUco markers are thus better aligned with the requirements of our task and are well integrated in OpenCV for fast multi-marker detection on low-cost hardware. As reported by Garrido-Jurado et al. [5], ArUco provides robust detection of binary square fiducial markers and is suitable for real-time applications. To support tactile identification, each tangible uses a split top surface: one part contains the ArUco marker for camera recognition, and the other contains 8-dot Computer Braille printed on transparent foil. Compared with traditional 6-dot Braille, the 8-dot format allows a more compact encoding of single digits and operators.

Grid Early prototypes using microcapsule paper and magnetic tangibles caused issues with unintended magnetic attraction and repulsion, as well as accidental displacement. Consequently, the final design uses a laser-cut thin wooden board. Each cell is a cutout with a diameter slightly larger than the base of the tangible, creating passive haptic constraints. This physical slotting mechanism stabilizes the tangibles without magnets while facilitating easy removal. Users can perceive the grid structure by touching the cutouts (see Fig. 1(b)) and identify the row and column coordinates via Braille labels on the edges of the board.

3.3 Interaction and Voice Assistance

The core interaction follows a "Prompt-Action-Feedback" loop. The system guides the user by providing a task prompt to set the context (e.g., “Set up the operands”), followed by specific action instructions (e.g., “Place number 2 at position B2”). The user then places a tangible, and the system verifies the position (Row, Col) and ID via computer vision. This allows for real-time error correction or confirmation.

To reduce cognitive load, a voice assistant is integrated, activated by the wake word "Hello Assistant". It supports four intent categories:

- Task Prompting: "What is the next step?"
- Action Instruction: "What is the next action"
- Contextual Deictic Query: Users point to a tangible and ask, “What is this?”, and the system identifies the digit and its place-value context.
- Calculation Support: Queries such as "What is 5 times 9?" serve to reduce the mental arithmetic load.

The system is designed for robust interruptibility. User queries can interrupt TTS output at any time. Once the Q&A is completed, the system automatically resumes the previous state to ensure continuity. Similarly, if a navigation task is interrupted, it is automatically restarted.

Table 1. Participant demographics. G: Gender (M=male, F=female), Age: in years, Hand: Handedness (R=Right, L=Left, B=Both), VA: Visual Acuity (TB = totally blind, LB = legally blind, VI = visually impaired), VL: Vision Loss (Congenital, Baby (≤ 3), Child (4–13), Adolescent (≥ 14)), BRP: Braille Reading Proficiency, Calculation Method: Daily calculation habits

P	Age	G	Hand	VA	VL	BRP	Calculation Method
P1	2004	M	B	TB	Congenital	Advanced	Mental/Braille/App
P2	1995	F	R	VI	Congenital	None	Mental
P3	2000	M	B	VI	Adolescent	Basic	Mental/Braille
P4	2008	F	R	TB	Child	Advanced	Calculator/App
P5	2007	F	B	TB	Congenital	Advanced	Calculator/App
P6	1998	F	R	TB	Congenital	Advanced	App
P7	2005	F	L	TB	Congenital	Advanced	Calculator/App
P8	2007	M	R	TB	Child	Advanced	Audio/App

4 User Study

We conducted a user study at the Educational Centre for the Blind and Visually Impaired (BBS Nürnberg) with eight participants who are blind or have low vision. Participant demographics, including their visual acuity and preferred calculation methods, are detailed in Table 1. The experimental task consisted of standard written arithmetic operations (e.g., 23×45), performed using the tangible grid layout. We used a within-subjects design with three experimental conditions:

- V1 (Baseline): The system provided step-by-step task prompts, action instructions, and verification feedback. Users relied solely on tactile exploration for spatial orientation on the grid.
- V2 (Voice Assistant): V1 plus voice-based Q&A, but without active finger navigation.
- V3 (Full Support): V2 plus finger navigation. Before each action, the system guided the user's finger to the target cell using clock-face directional cues (e.g., "9 o'clock") and sonification, where the pitch frequency increased as the finger approached the target.

We collected both quantitative and qualitative data, including demographic questionnaires and post-task evaluations. Subjective workload was measured using the NASA-TLX, and usability was assessed via the System Usability Scale (SUS). Additionally, custom 5-point Likert scales were used to evaluate the perceived effectiveness of the guidance and feedback mechanisms.

5 Results

The overall usability of the system was evaluated using the System Usability Scale (SUS). As shown in Fig. 2, individual scores ranged from 57.5 to 95.0 with

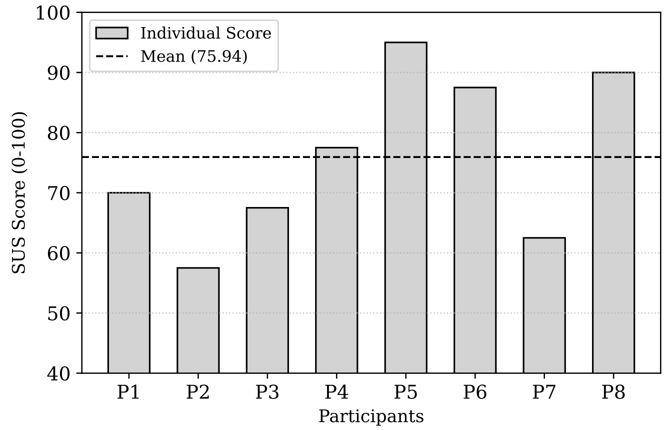


Fig. 2. Individual System Usability Scale (SUS) scores for the eight participants. The dashed line represents the mean score.

a mean score of 75.9 (SD=13.8). According to Bangor et al. [2], this falls into the “Good” range (Grade B).

Because SUS reflects overall usability rather than mathematics learning outcomes, we complemented it with custom Likert ratings and qualitative feedback on the physical layer. Participants rated the grid structure as easy to understand (4.50) and the cell boundaries as distinguishable (4.63). This interpretation is supported by qualitative feedback: P7 noted that the laser-cut holes were more stable than magnets, and P1 reported that placing one or two tangibles was enough to understand the basic interaction.

At the same time, although “the intuitiveness of the tangibles” (4.13) and “ease of manipulation” (4.38) were also rated positively, our observations showed that users still relied strongly on the Braille labels at the grid edges for row and column orientation. Once their fingers moved into the middle area, some participants became more likely to lose orientation because continuous physical reference points were missing. In addition, a few participants, such as P7 who preferred the left hand, showed some difficulty with a navigation approach that mainly designed for right-hand use.

Subjective workload was analyzed using the NASA-TLX (see Fig. 3). V1 (Baseline) and V2 (Voice Assistant) showed similar workload levels across most dimensions, with V2 yielding slightly more favorable scores in Mental Demand (M=2.38) and Performance (inverted score M=1.25). This suggests that the voice assistant provided some support.

By contrast, V3 (Full Support) showed higher workload ratings across all dimensions, especially in Temporal Demand (M = 2.00), Effort (M = 3.13), and Frustration (M = 2.50). Qualitative feedback indicates that system latency was the main cause of this additional burden. P1 explained that, despite the

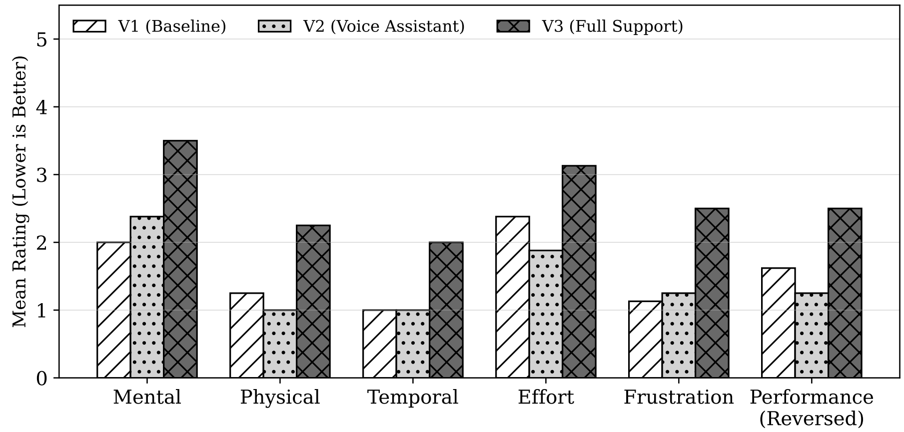


Fig. 3. Mean NASA-TLX scores across six dimensions for the three conditions: V1 (Baseline), V2 (Voice Assistant), and V3 (Full Support). Note: The "Performance" scale is inverted, so that for all dimensions, a lower bar indicates a better user experience.

navigation cues, it was still difficult to confirm the correct position. The finger often overshot the target and then circled around it, for example receiving "9 o'clock," then "6 o'clock," and then "3 o'clock," because the audio feedback lagged behind the hand movement. P3 likewise described the continuous beeping as "annoying" because of the delay. At the same time, several participants reported that they gradually adapted to the latency after a few attempts. When they deliberately slowed down their finger movements, the navigation function became usable. This suggests that the main issue in V3 may have been hardware-related latency rather than the interaction concept itself.

The higher Mental Demand in V3 (M=3.50) was also related to spatial cue interpretation. Three congenitally blind participants reported that clock-face directions were not an intuitive reference and first had to be translated into left/right directions, adding cognitive effort.

6 Discussion and Outlook

Standard arithmetic tasks are typically represented and calculated through a two-dimensional spatial layout, while non-visual access is often sequential, for example through screen readers. Drag & Speak addresses this gap by providing a tangible grid as a stable physical anchor and by translating spatial arithmetic into a structured sequence of tactile and auditory cues. This enables users to explore rows and columns step by step and supports access to spatial relations in a more manageable way.

Limitations have been detected in automatically starting to guide the user automatically. Controllability of the interaction can be enhanced in future versions.

Future work will focus on adapting the system for use in primary school education and connecting it more closely to early mathematics teaching. In particular, future versions could support foundational concepts such as place value and carrying, which can be difficult to access in non-visual learning contexts. In this way, Drag & Speak may develop into not only an assistive system, but also a pedagogical tool for inclusive mathematics education.

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